

An Instructor Ratings Analysis: How Relationships Among Ratings and Factors Associated with Teaching Quality Based

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1 Introduction

Teaching isn't just about a professor lecturing; it's also a journey of improvement, guided by feedback from students. Therefore, student evaluation feedback is highly valued for university to assess teaching quality. However, ratings may be related to various teaching-related factors. This study utilizes student evaluation data from a large American university to explore the relationships among the five rating dimensions and analyze the factors influencing the overall teaching quality of the professor.

2 Relationships Among the Ratings

Overall, STEM instructors received the lowest mean ratings across all five dimensions, particularly in easiness (2.87), suggesting that students who enrolled in STEM courses have relatively higher difficulty and are less satisfied with the quality of the courses. In contrast, Pre-prof instructors received the highest mean ratings in quality, helpfulness, and easiness, but their average student interest rating was relatively low with level of only 3.23. It may suggest that higher instructor quality ratings do not necessarily correspond to greater student interest in the field.

Notably, the standard deviations were similar across the four disciplines, which suggest students tend to come out with similar conclusion. But for easiness, it showed more noticeable variation, which may indicate that the level of difficulty of the subject tend to be vary.

As shown in Figure 1, all five rating dimensions were positively correlated. Quality was strongly associated with helpfulness ($r = 0.981$) and clarity ($r = 0.976$), suggesting that these dimensions reflect closely related aspects of teaching evaluations. In contrast, quality showed weaker correlations with easiness ($r = 0.565$) and student interest ($r = 0.471$). The associations between these rating dimensions and overall instructor quality will be further examined in the subsequent regression analysis.

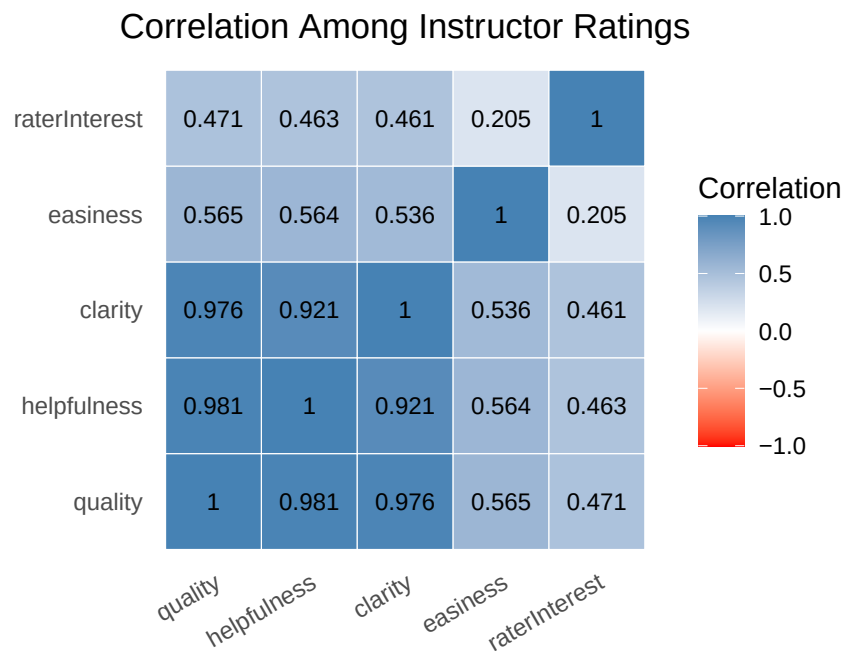


Figure 1

3 Model

We first examined the distributions of numerical variables using histograms and found that quality was slightly left-skewed, while easiness and raterInterest were relatively symmetric. Besides, numYears showed a strong concentration at its maximum of 11 years. We then used scatterplots to explore their relationships with quality. Both easiness and raterInterest showed generally positive and approximately linear relationships, whereas we do not observe a clear pattern between numYears and quality. Based on these observations, we decided to keep the variables on their original scales for the initial regression model(model 1) and further assess their suitability through residual diagnostics.

After performing diagnostic plots, the results suggest that the model1 performs reasonably well, with no major concerns about linearity or normality. Although the Scale-Location plot shows slightly spreaded, but the Breusch-Pagan test found no significant evidence of heteroscedasticity with $p = 0.133$. We also identified 14 potentially influential observations using Cook's distance, however, sensitivity analysis showed that the main regression coefficients remained relatively stable after excluding them. Meanwhile, the residuals showed no clear pattern across observation order, and VIF values close to 1. So multicollinearity and independent are satisfied. Based on these findings, we decided to retain the original model without further adjustments.

$$\begin{aligned} \text{Quality}_i \mid \mu_i, \sigma &\sim \text{Normal}(\mu_i, \sigma) \\ \mu_i &= \beta_0 + \beta_1 \cdot \text{gender}_i + \beta_2 \cdot \text{numYears}_i \\ &\quad + \beta_3 \cdot \text{PreProf}_i + \beta_4 \cdot \text{SocSci}_i + \beta_5 \cdot \text{STEM}_i \\ &\quad + \beta_6 \cdot \text{easiness}_i + \beta_7 \cdot \text{raterInterest}_i + \beta_8 \cdot \text{pepper}_i \end{aligned}$$

4 Statistical Significance and Practical Importance

As shown in Table 1, easiness, student interest,pepper, and STEM discipline were positively associated with instructor quality after holding for other predictors (all $p < 0.001$). The most impactable variable is Pepper yes or no, students tend to give a higher rating in the term of any other condition. statistically, Holding other variables constant, a one-point increase in student interest was associated with approximately 0.54 higher quality ratings. Although STEM instructors received the lowest average quality ratings, but after adjusting for other predictors, STEM instructors were estimated to receive quality ratings approximately 0.30 points higher than Humanities instructors. In contrast, the rest variable showed no statistically significant associations. The confidence intervals for significant predictors excluded zero, which support null hypothesis which they are positively associated.

Table 1: Multiple Linear Regression Results for Instructor Quality

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.181	0.242	[-0.657, 0.296]	0.457
gender	0.047	0.065	[-0.081, 0.174]	0.472
numYears	0.018	0.010	[-0.002, 0.037]	0.081
disciplinePre-prof	0.097	0.091	[-0.083, 0.276]	0.291
disciplineSocSci	0.019	0.089	[-0.156, 0.193]	0.834
disciplineSTEM	0.295	0.081	[0.135, 0.455]	<0.001
easiness	0.513	0.042	[0.429, 0.596]	<0.001
raterInterest	0.544	0.059	[0.427, 0.661]	<0.001
pepper	0.562	0.099	[0.366, 0.757]	<0.001

5 Quality and Discipline

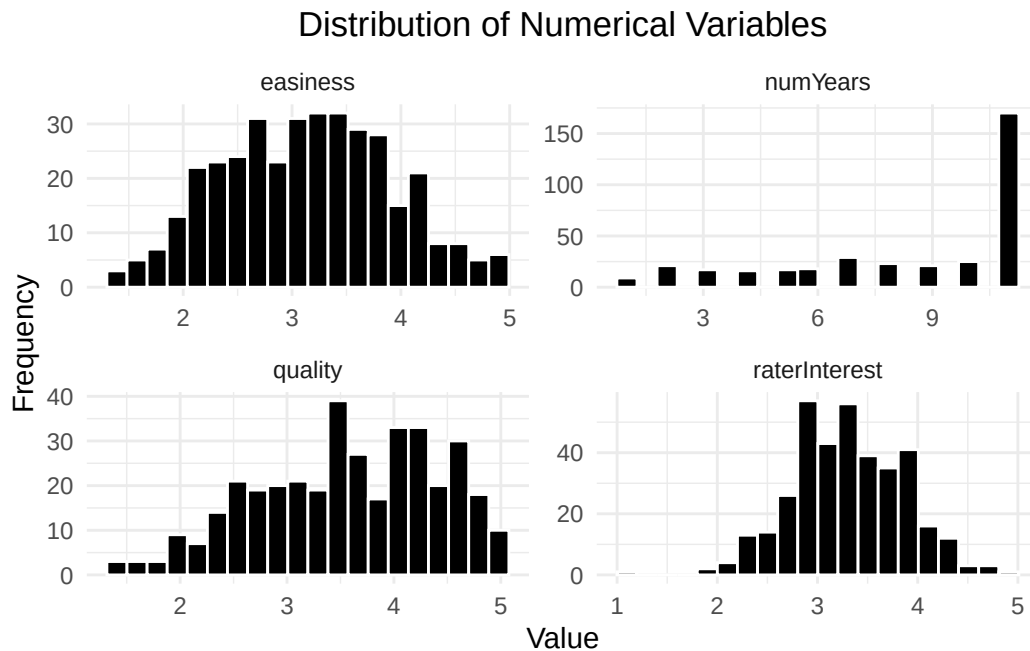
A Appendix 1

Table 2: Instructor Ratings by Discipline (Mean and SD)

Discipline	N	Quality	Helpfulness	Clarity	Easiness	Rater Interest
Hum	134	3.58 (0.85)	3.62 (0.86)	3.54 (0.84)	3.19 (0.75)	3.41 (0.55)
Pre-prof	63	3.62 (0.86)	3.68 (0.89)	3.57 (0.87)	3.33 (0.86)	3.23 (0.58)
STEM	103	3.53 (0.82)	3.62 (0.80)	3.43 (0.88)	2.87 (0.73)	3.15 (0.52)
SocSci	66	3.61 (0.84)	3.62 (0.84)	3.59 (0.87)	3.25 (0.72)	3.44 (0.51)

```
#histogram
variables <- c("quality", "numYears", "easiness", "raterInterest")
dist_data <- data %>%
  pivot_longer(all_of(variables), names_to = "Variable", values_to = "Value")

# let review distributions
ggplot(dist_data, aes(x = Value)) +
  geom_histogram(bins = 20, fill = "black", color = "white") +
  facet_wrap(~Variable, scales = "free") +
  labs(title = "Distribution of Numerical Variables", x = "Value", y = "Frequency") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))
```



```
scatter_data <- data[, c("quality", "numYears", "easiness", "raterInterest")]

scatter_long <- pivot_longer(scatter_data, -quality,
                             names_to = "Variable", values_to = "Value")

# Plot relationships with quality
ggplot(scatter_long, aes(x = Value, y = quality)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  facet_wrap(~Variable, scales = "free_x") +
  labs(title = "Relationships Between Numerical Predictors and Quality",
       x = "Predictor Value", y = "Quality") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))
```

`geom\_smooth()` using formula = 'y ~ x'

